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Avalue

Board Hardware I/O Driver and Library API Platform Programmer's Reference Manual

Rev. 3.00.06.16

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1. API Programmer's Reference

※ Namespace: Avalue

1.1 Avalue API

Avalue product information and operation.

Syntax		Description
Managed	C++	Avalue::AvalueAPI
	C#	Avalue.AvalueAPI

1.1.1 Enumerations

1.1.1.1 ACCESS

Basic read/write access.

Member Name	Value	Description
IOPORT	0	I/O port index/data
MEMORY	1	Memory address

1.1.1.2 RWSIZE

Data read/write size.

Member Name	Value	Description
BYTE	0	8 bit
WORD	1	16 bit
DWORD	2	32 bit

1.1.2 Version

1.1.2.1 Number

Avalue API version number.

Syntax		Description
Managed	C++	unsigned long AvalueAPI::Version
	C#	uint AvalueAPI.Version
Unmanaged		unsigned long AvalueAPI_Version()

Return Value:

For example: 0x03000000.

1.1.2.2 Text

Avalue API version text.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::Version_Text
	C#	string AvalueAPI.Version_Text
Unmanaged		wchar_t* AvalueAPI_Version_Text()

Return Value:

For example: 3.00.00.00.

1.1.3 Start

Open Avalue driver and initialize Avalue API.

Syntax		Description
Managed	C++	bool AvalueAPI::Start()
	C#	bool AvalueAPI.Start()
Unmanaged		bool AvalueAPI_Start()

Return Value	Description
FALSE	Driver not installed or driver not opened
TRUE	Success

Remarks:

Call this method first before use all class methods.

1.1.4 Stop

Close Avalue driver and disable Avalue API.

Syntax		Description
Managed	C++	bool AvalueAPI::Stop()
	C#	bool AvalueAPI.Stop()
Unmanaged		bool AvalueAPI_Stop()

Return Value	Description
FALSE	API not started
TRUE	Success

Remarks:

Call this method if no longer to use all class methods.

1.1.5 Read

Read data.

Syntax		Description
Managed	C++	unsigned long AvalueAPI::Read(AvalueAPI::ACCESS access, AvalueAPI::RWSIZE size, unsigned long command)
	C#	uint AvalueAPI.Read(AvalueAPI.ACCESS access, AvalueAPI.RWSIZE size, uint command)
Unmanaged		unsigned long AvalueAPI_Read(ACCESS access, RWSIZE size, unsigned long command)

Parameters	Description
access	Refer to 1.1.1.1 ACCESS
size	Refer to 1.1.1.2 RWSIZE
command	I/O port index or memory address

Return Value	Description
0xFFFFFFFF	Class not initialized or parameters incorrect
0x00000000 ~ 0xFFFFFFFF	Return data

1.1.6 Write

Write data.

Syntax		Description
Managed	C++	bool AvalueAPI::Write(AvalueAPI::ACCESS access, AvalueAPI::RWSIZE size, unsigned long command, unsigned long data)
	C#	bool AvalueAPI.Write(AvalueAPI.ACCESS access, AvalueAPI.RWSIZE size, uint command, uint data)
Unmanaged		bool AvalueAPI_Write(ACCESS access, RWSIZE size, unsigned long command, unsigned long data)

Parameters	Description
access	Refer to 1.1.1.1 ACCESS
size	Refer to 1.1.1.2 RWSIZE
command	I/O port index or memory address
data	Write data

Return Value	Description
FALSE	Class not initialized or parameters incorrect
TRUE	Success

1.2 Info

Motherboard information.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::Info
	C#	Avalue.AvalueAPI.Info

1.2.1 Enumerations

1.2.1.1 PWR

Power mode.

Member Name	Value	Description
NA	-1	API not started or method not supported
AT	0	AT mode
ATX	1	ATX mode

1.2.2 Struct

1.2.2.1 NET

Network detail.

Member Name		Description
C++	C#	
System::String^ Name	string Name	Network name
System::String^ MAC	string MAC	Network MAC address

1.2.3 DMI_SystemManufacturer

BIOS DMI information - System Manufacturer.

Syntax		Description
Managed	C++	System::String^ Avalue::AvalueAPI::Info::DMI_SystemManufacturer()
	C#	string Avalue.AvalueAPI.Info.DMI_SystemManufacturer()
Unmanaged		wchar_t* DMI_SystemManufacturer()

Remarks:

1.1.3 Start is not required.

1.2.4 DMI_SystemProduct

BIOS DMI information - System Product.

Syntax		Description
Managed	C++	System::String^ Avalue::AvalueAPI::Info::DMI_SystemProduct()
	C#	string Avalue.AvalueAPI.Info.DMI_SystemProduct()
Unmanaged		wchar_t* DMI_SystemProduct()

Remarks:

1.1.3 Start is not required.

1.2.5 DMI_SystemSerialNumber

BIOS DMI information - System Serial Number.

Syntax		Description
Managed	C++	System::String^ Avalue::AvalueAPI::Info::DMI_SystemSerialNumber()
	C#	string Avalue.AvalueAPI.Info.DMI_SystemSerialNumber()
Unmanaged		wchar_t* DMI_SystemSerialNumber()

Remarks:

1.1.3 Start is not required.

1.2.6 DMI_SystemSKUNumber

BIOS DMI information - System SKU Number.

Syntax		Description
Managed	C++	System::String^ Avalue::AvalueAPI::Info::DMI_SystemSKUNumber()
	C#	string Avalue.AvalueAPI.Info.DMI_SystemSKUNumber()
Unmanaged		wchar_t* DMI_SystemSKUNumber()

Remarks:

1.1.3 Start is not required.

1.2.7 DMI_BaseboardVersion

BIOS DMI information - Baseboard Version.

Syntax		Description
Managed	C++	System::String^ Avalue::AvalueAPI::Info::DMI_BaseboardVersion()
	C#	string Avalue.AvalueAPI.Info.DMI_BaseboardVersion()
Unmanaged		wchar_t* DMI_BaseboardVersion()

Remarks:

1.1.3 Start is not required.

1.2.8 DMI_BaseboardSerialNumber

BIOS DMI information - Baseboard Serial Number.

Syntax		Description
Managed	C++	System::String^ Avalue::AvalueAPI::Info::DMI_BaseboardSerialNumber()
	C#	string Avalue.AvalueAPI.Info.DMI_BaseboardSerialNumber()
Unmanaged		wchar_t* DMI_BaseboardSerialNumber()

Remarks:

1.1.3 Start is not required.

1.2.9 Manufacturer

Motherboard manufacturer.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::Info::Manufacturer()
	C#	string AvalueAPI.Info.Manufacturer()
Unmanaged		wchar_t* Manufacturer()

Remarks:

1.1.3 Start is not required.

1.2.10 Motherboard

Motherboard name.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::Info::Motherboard()
	C#	string AvalueAPI.Info.Motherboard()
Unmanaged		wchar_t* Motherboard()

Remarks:

1.1.3 Start is not required.

1.2.11 BIOS Version

BIOS version.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::Info::BIOS_Version()
	C#	string AvalueAPI.Info.BIOS_Version()
Unmanaged		wchar_t* BIOS_Version()

Remarks:

1.1.3 Start is not required.

1.2.12 EC Version

EC version.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::Info::EC_Version()
	C#	string AvalueAPI.Info.EC_Version()
Unmanaged		wchar_t* EC_Version()

Return Value	Description
N/A	API not started or method not supported
Others	Return version

1.2.13 NPU Version

NPU version.

Syntax		Description
Managed	C++	System::String^ Avalue::AvalueAPI::Info::NPU_Version()
	C#	string Avalue.AvalueAPI.Info.NPU_Version()

Return Value	Description
N/A	API not started or method not supported
Others	Return version

1.2.14 IPMI

1.2.14.1 Device IP

Device IP.

Syntax		Description
Managed	C++	System::String^ Avalue::AvalueAPI::Info::IPMI_DeviceIP()
	C#	string Avalue.AvalueAPI.Info.IPMI_DeviceIP()

Return Value	Description
N/A	API not started or method not supported
Others	Return IP address

1.2.14.2 Device MAC

Device MAC.

Syntax		Description
Managed	C++	System::String^ Avalue::AvalueAPI::Info::IPMI_DeviceMAC()
	C#	string Avalue.AvalueAPI.Info.IPMI_DeviceMAC()

Return Value	Description
N/A	API not started or method not supported
Others	Return MAC address

1.2.15 CPU

1.2.15.1 Name

CPU name.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::Info::CPU_Name()
	C#	string AvalueAPI.Info.CPU_Name()
Unmanaged		wchar_t* CPU_Name()

Remarks:

1.1.3 Start is not required.

1.2.15.2 Identifier

CPU identifier.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::Info::CPU_Identifier()
	C#	string AvalueAPI.Info.CPU_Identifier()
Unmanaged		wchar_t* CPU_Identifier()

Remarks:

1.1.3 Start is not required.

1.2.15.3 Microcode

CPU microcode.

Syntax		Description
Managed	C++	unsigned long AvalueAPI::Info::CPU_Microcode()
	C#	uint AvalueAPI.Info.CPU_Microcode()
Unmanaged		unsigned long CPU_Microcode()

Remarks:

1.1.3 Start is not required.

1.2.15.4 Check

Check whether the current CPU meets the SA-00088 security requirements.

Syntax		Description
Managed	C++	bool AvalueAPI::Info::CPU_Check()
	C#	bool AvalueAPI.Info.CPU_Check()
Unmanaged		bool CPU_Check()

Return Value	Description
FALSE	CPU does not meet the requirements. Possible reasons include: - API not started - Does not meet SA-00088 - End-of-Servicing-Lifetime (EOSL) reached
TRUE	CPU meets the SA-00088 requirements. Check passed.

References:

- Intel SA-00088 Security Advisory

<https://www.intel.com/content/www/us/en/security-center/advisory/intel-sa-00088.html>

- Intel Affected Processors Table

<https://www.intel.com/content/www/us/en/developer/topic-technology/software-security-guidance/processors-affected-consolidated-product-cpu-model.html>

- Intel GitHub CSV List

https://github.com/intel/Intel-affected-processor-list/blob/main/Intel_affected_processor_list.csv

1.2.16 Power Mode

Power mode.

Syntax		Description
Managed	C++	AvalueAPI::Info::PWR AvalueAPI::Info::PowerMode()
	C#	AvalueAPI.Info.PWR AvalueAPI.Info.PowerMode()
Unmanaged		PWR PowerMode()

Return Value:

Refer to [1.2.1.1 PWR](#).

1.2.17 Count Times

Counts for each system power state. Feature availability may vary by product.

Syntax		Description
Managed	C++	int Avalue::AvalueAPI::Info::STATS_CountTimesS0()
		int Avalue::AvalueAPI::Info::STATS_CountTimesG3()
		int Avalue::AvalueAPI::Info::STATS_CountTimesS5()
	C#	int Avalue.AvalueAPI.Info.STATS_CountTimesS0() int Avalue.AvalueAPI.Info.STATS_CountTimesG3() int Avalue.AvalueAPI.Info.STATS_CountTimesS5()

Return Value	Description
-1	API not started or method not supported
Others	Entry count

1.2.18 Count Times Clear

Clear counts for each system power state. Feature availability may vary by product.

Syntax		Description
Managed	C++	bool Avalue::AvalueAPI::Info::STATS_CountTimesClearS0()
		bool Avalue::AvalueAPI::Info::STATS_CountTimesClearG3()
		bool Avalue::AvalueAPI::Info::STATS_CountTimesClearS5()
	C#	bool Avalue.AvalueAPI.Info.STATS_CountTimesClearS0() bool Avalue.AvalueAPI.Info.STATS_CountTimesClearG3() bool Avalue.AvalueAPI.Info.STATS_CountTimesClearS5()

Return Value	Description
FALSE	API not started or method not supported
TRUE	Success

1.2.19 Duration

Duration per system power state. Feature availability may vary by product.

Syntax		Description
Managed	C++	long long Avalue::AvalueAPI::Info::STATS_DurationS0()
	C#	long Avalue.AvalueAPI.Info.STATS_DurationS0()

Return Value	Description
-1	API not started or method not supported
Others	Total duration in seconds

1.2.20 Duration Clear

Clear duration per system power state. Feature availability may vary by product.

Syntax		Description
Managed	C++	bool Avalue::AvalueAPI::Info::STATS_DurationClearS0()
	C#	bool Avalue.AvalueAPI.Info.STATS_DurationClearS0()

Return Value	Description
FALSE	API not started or method not supported
TRUE	Success

1.2.21 Network

Network information.

Syntax		Description
Managed	C++	array<AvalueAPI::Info::NET>^ AvalueAPI::Info::Network()
	C#	AvalueAPI.Info.NET[] AvalueAPI.Info.Network()
Unmanaged		wchar_t** Network(int* size)

Return Value:

Refer to [1.2.2.1 NET](#).

Remarks:

[1.1.3 Start](#) is not required.

1.3 Setting

Motherboard setting.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::Seting
	C#	Avalue.AvalueAPI.Setting

1.3.1 Enumerations

1.3.1.1 TYPE

COM port type.

Member Name	Value	Description
NA	-1	Class not initialized or method not supported
RS232	0	RS-232
RS422	1	RS-422
RS485	2	RS-485

1.3.1.2 STATE

Function state.

Member Name	Value	Description
NA	-1	Class not initialized or method not supported
DISABLE	0	Disable
ENABLE	1	Enable

1.3.1.3 RES

RTC result.

Member Name	Value	Description
NA	-1	Class not initialized or method not supported
OK	0	RTC check passed
INVALID_YEAR	1	RTC checked, year invalid
FROZEN	2	RTC freeze handled
FROZEN_INVALID_YEAR	3	RTC freeze handled, year invalid

1.3.2 Struct

1.3.2.1 TIME

RTC time.

Member Name	Description
int Year	RTC year value (2000–2099)
int Month	RTC month value (1–12)
int Day	RTC day value (1–31)
int Hour	RTC hour value (0–23)
int Minute	RTC minute value (0–59)
int Second	RTC second value (0–59)

1.3.3 AMP Gain

1.3.3.1 Get

Get onboard AMP gain.

Syntax		Description
Managed	C++	int AvalueAPI::Setting::AMP_Gain()
	C#	int AvalueAPI.Setting.AMP_Gain()
Unmanaged		int AMP_Gain_Get()

Return Value	Description
-1	Class not initialized or method not supported
0 ~ 3	Refer to 1.3.2.3 Text

1.3.3.2 Set

Set onboard AMP gain.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::AMP_Gain(int value)
	C#	bool AvalueAPI.Setting.AMP_Gain(int value)
Unmanaged		bool AMP_Gain_Set(int value)

Parameters	Description
value	0 ~ 3

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.3.3.3 Text

Get onboard AMP gain value text.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::Setting::AMP_Gain_Text(int value)
	C#	string AvalueAPI.Setting.AMP_Gain_Text(int value)
Unmanaged		wchar_t* AMP_Gain_Text(int value)

Parameters	Description
value	0 ~ 3

Return Value	Description
N/A	Class not initialized, method not supported or parameters incorrect
Others	The dB text

1.3.4 COM Type

1.3.4.1 Get

Get COM port type.

Syntax		Description
Managed	C++	AvalueAPI::Setting::TYPE AvalueAPI::Setting::COM_Type(int number)
	C#	AvalueAPI.Setting.TYPE AvalueAPI.Setting.COM_Type(int number)
Unmanaged		TYPE COM_Type_Get(int number)

Parameters	Description
number	0 ~ 2

Return Value:

Refer to [1.3.1.1 TYPE](#).

1.3.4.2 Set

Set COM port type.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::COM_Type(int number, AvalueAPI::Setting::TYPE type)
	C#	bool AvalueAPI.Setting.COM_Type(int number, AvalueAPI.Setting.TYPE type)
Unmanaged		bool COM_Type_Set(int number, TYPE type)

Parameters	Description
number	0 ~ 3
type	Refer to 1.3.1.1 TYPE

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.3.5 Touch

1.3.5.1 Get

Get onboard touch state.

Syntax		Description
Managed	C++	AvalueAPI::Setting::STATE AvalueAPI::Setting::Touch()
	C#	AvalueAPI.Setting.STATE AvalueAPI.Setting.Touch()
Unmanaged		STATE Touch_Get()

Return Value:

Refer to [1.3.1.2 STATE](#).

1.3.5.2 Set

Set onboard touch state.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::Touch(AvalueAPI::Setting::STATE state)
	C#	bool AvalueAPI.Setting.Touch(AvalueAPI.Setting.STATE state)
Unmanaged		bool Touch_Set(STATE state)

Parameters	Description
state	Refer to 1.3.1.2 STATE

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.3.6 USB Standby Power

1.3.6.1 Get

Get onboard USB standby power state.

Syntax		Description
Managed	C++	AvalueAPI::Setting::STATE AvalueAPI::Setting::USB_Standby_Power(int number)
	C#	AvalueAPI.Setting.STATE AvalueAPI.Setting.USB_Standby_Power(int number)
Unmanaged		STATE USB_Standby_Power_Get(int number)

Parameters	Description
number	0 ~ 5

Return Value:

Refer to [1.3.1.2 STATE](#).

1.3.6.2 Set

Set onboard USB standby power state.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::USB_Standby_Power(int number, AvalueAPI::Setting::STATE state)
	C#	bool AvalueAPI.Setting.USB_Standby_Power(int number, AvalueAPI.Setting.STATE state)
Unmanaged		bool USB_Standby_Power_Set(int number, STATE state)

Parameters	Description
number	0 ~ 5
state	Refer to 1.3.1.2 STATE

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.3.7 Buzzer

1.3.7.1 Mute

Set buzzer mute.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::BuzzerMute()
	C#	bool AvalueAPI.Setting.BuzzerMute()
Unmanaged		bool BuzzerMute()

Return Value	Description
FALSE	API not started
TRUE	Success

1.3.7.2 Beep

Set buzzer beep.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::BuzzerBeep(int hz) bool AvalueAPI::Setting::BuzzerBeep(int hz, int ms)
	C#	bool AvalueAPI.Setting.BuzzerBeep(int hz) bool AvalueAPI.Setting.BuzzerBeep(int hz, int ms)
Unmanaged		bool BuzzerBeep(int hz) bool BuzzerBeepTime(int hz, int ms)

Parameters	Description
hz	Beep frequency
ms	Beep time

Return Value	Description
FALSE	API not started or parameters incorrect
TRUE	Success

1.3.8 RTC

1.3.8.1 Check

Check RTC status.

Syntax		Description
Managed	C++	AvalueAPI::Setting::RESULT Avalue::AvalueAPI::Setting::RTC_Check() AvalueAPI::Setting::RESULT Avalue::AvalueAPI::Setting::RTC_Check(int min, int max)
	C#	AvalueAPI.Setting.RESULT Avalue.AvalueAPI.Setting.RTC_Check() AvalueAPI.Setting.RESULT Avalue.AvalueAPI.Setting.RTC_Check(int min, int max)

Parameters	Description
min	The year range minimum
max	The year range maximum

Return Value:

See section [1.3.1.3 RES](#) for details.

Remarks:

Checks the RTC status and validates the current year.

The default valid year range is from the API build year (e.g., 2025) to build year + 3.

A custom range can be defined through parameters.

1.3.8.2 Get

Get RTC time.

Syntax		Description
Managed	C++	AvalueAPI::Setting::TIME Avalue::AvalueAPI::Setting::RTC()
	C#	AvalueAPI.Setting.TIME Avalue.AvalueAPI.Setting.RTC()

Return Value:

See section [1.3.2.1 TIME](#) for details.

1.3.8.3 Set

Set RTC time.

Syntax		Description
Managed	C++	bool Avalue::AvalueAPI::Setting::RTC(AvalueAPI::Setting::TIME time)
	C#	bool Avalue.AvalueAPI.Setting.RTC(AvalueAPI.Setting.TIME time)

Parameters	Description
time	See section 1.3.2.1 TIME for details

Return Value	Description
FALSE	API not started or parameters incorrect
TRUE	Success

1.3.8.4 Sync

Synchronize RTC time to the Windows system.

Syntax		Description
Managed	C++	bool Avalue::AvalueAPI::Setting::RTC_SyncToOS()
	C#	bool Avalue.AvalueAPI.Setting.RTC_SyncToOS()

Return Value	Description
FALSE	API not started
TRUE	Success

Remarks:

This method requires administrator privileges to execute.

1.4 Cash Drawer

For ACP-BYTPOS, RITY153.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::Seting::CashDrawer
	C#	Avalue.AvalueAPI.Setting.CashDrawer

1.4.1 Enumerations

1.4.1.1 DRAWERSTATE

Cash drawer state.

Member Name	Value	Description
NA	-1	Class not initialized or method not supported
OPEN	0	Cash drawer opened
CLOSE	1	Cash drawer closed

1.4.2 Init

Cash drawer class initialize.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::CashDrawer::Init()
	C#	bool AvalueAPI.Setting.CashDrawer.Init()

Return Value	Description
FALSE	API not started or class not supported
TRUE	Success

Remarks:

Call this method first before use CashDrawer class methods.

1.4.3 Disable

Cash drawer class disable.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::CashDrawer::Disable()
	C#	bool AvalueAPI.Setting.CashDrawer.Disable()

Return Value	Description
FALSE	Class not initialized
TRUE	Success

Remarks:

Call this method if no longer to use CashDrawer class methods.

1.4.4 Kick Out

Kick out.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::CashDrawer::KickOut1() bool AvalueAPI::Setting::CashDrawer::KickOut2()
	C#	bool AvalueAPI.Setting.CashDrawer.KickOut1() bool AvalueAPI.Setting.CashDrawer.KickOut2()

Return Value	Description
FALSE	Class not initialized or method not supported
TRUE	Success

1.4.5 Sense

Cash drawer sense.

Syntax		Description
Managed	C++	AvalueAPI::Setting::CashDrawer::DRAWERSTATE AvalueAPI::Setting::CashDrawer::Sense()
	C#	AvalueAPI.Setting.CashDrawer.DRAWERSTATE AvalueAPI.Setting.CashDrawer.Sense()

Return Value:

Refer to **1.4.1.1 DRAWERSTATE**.

1.5 VMS

VMS Series.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::Seting::VMS
	C#	Avalue.AvalueAPI.Setting.VMS

1.5.1 Enumerations

1.5.1.1 RANGE

Input voltage range.

Member Name	Value	Description
V24	0	24V
V12	1	12V
UNKNOWN	2	Unknown
V9_36	3	9V ~ 36V

1.5.1.2 SWITCH

Ignition switch.

Member Name	Value	Description
NA	-1	Class not initialized or method not supported
OFF	0	Off
ON	1	On

1.5.1.3 TRIGGER

Vin boot/shutdown trigger.

Member Name	Value	Description
NA	-1	Class not initialized or method not supported
DISABLE	0	Disable
T1	1	11.5V, 10.5V / 23V, 21V
T2	2	12.0V, 11.0V / 24V, 22V
T3	3	12.5V, 11.0V / 25V, 22V
T4	4	12.5V, 11.5V / 25V, 23V

1.5.1.4 TIME

Delay time.

Member Name	Value	Description
DISABLE	0	Disable
S10	1	10 sec
S30	2	30 sec
M1	3	1 min
M5	4	5 min
M10	5	10 min
M15	6	15 min
M30	7	30 min
H1	8	1 hour

1.5.2 Power Input

Get power input range.

Syntax		Description
Managed	C++	AvalueAPI::Setting::VMS::RANGE AvalueAPI::Setting::VMS::PowerInput()
	C#	AvalueAPI.Setting.VMS.RANGE AvalueAPI.Setting.VMS.PowerInput()

Return Value:

Refer to [1.5.1.1 RANGE](#).

1.5.3 Ignition

Get ignition.

Syntax		Description
Managed	C++	AvalueAPI::Setting::VMS::SWITCH AvalueAPI::Setting::VMS::Ignition()
	C#	AvalueAPI.Setting.VMS.SWITCH AvalueAPI.Setting.VMS.Ignition()

Return Value:

Refer to [1.5.1.2 SWITCH](#).

1.5.4 Vin Boot Shutdown

1.5.4.1 Get

Get Vin boot/shutdown trigger.

Syntax		Description
Managed	C++	AvalueAPI::Setting::VMS::TRIGGER AvalueAPI::Setting::VMS::VinBootShutdown()
	C#	AvalueAPI.Setting.VMS.TRIGGER AvalueAPI.Setting.VMS.VinBootShutdown()

Return Value:

Refer to [1.5.1.3 TRIGGER](#).

1.5.4.2 Set

Set Vin boot/shutdown trigger.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::VMS::VinBootShutdown(AvalueAPI::Setting::VMS::TRIGGER trigger)
	C#	bool AvalueAPI.Setting.VMS.VinBootShutdown(AvalueAPI.Setting.VMS.TRIGGER trigger)

Parameters	Description
trigger	Refer to 1.5.1.3 TRIGGER

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.5.5 Delay

1.5.5.1 On

Power on delay.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::VMS::DelayON(AvalueAPI::Setting::VMS::TIME time)
	C#	bool AvalueAPI.Setting.VMS.DelayON(AvalueAPI.Setting.VMS.TIME time)

Parameters	Description
time	Refer to 1.5.1.4 TIME

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.5.5.2 Off

Power off delay.

Syntax		Description
Managed	C++	bool AvalueAPI::Setting::VMS::DelayOFF(AvalueAPI::Setting::VMS::TIME time)
	C#	bool AvalueAPI.Setting.VMS.DelayOFF(AvalueAPI.Setting.VMS.TIME time)

Parameters	Description
time	Refer to 1.5.1.4 TIME

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.5.6 Power OFF Countdown

Get power off countdown.

Syntax		Description
Managed	C++	int AvalueAPI::Setting::VMS::PowerOFFCountdown()
	C#	int AvalueAPI.Setting.VMS.PowerOFFCountdown()

Return Value	Description
-1	Class not initialized or method not supported
Others	Countdown time

1.6 SMBUS

System management bus.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::SMBUS
	C#	Avalue.AvalueAPI.SMBUS

1.6.1 Enumerations

1.6.1.1 STATUS

SMBUS status.

Member Name	Value	Description
NA	-1	Class not initialized
READY	0	Ready
HBSY	1	Host busy
INTR	2	Interrupt
DERR	3	Device error
BERR	4	Bus error
FAIL	5	Failed
SMSTS	6	SMBALERT_STS
IUS	7	In use status

1.6.2 Init

SMBUS class initialize.

Syntax		Description
Managed	C++	bool AvalueAPI::SMBUS::Init()
	C#	bool AvalueAPI.SMBUS.Init()
Unmanaged		bool SMBUS_Init()

Return Value	Description
FALSE	API not started or class not supported
TRUE	Success

Remarks:

Call this method first before use SMBUS class methods.

1.6.3 Disable

SMBUS class disable.

Syntax		Description
Managed	C++	bool AvalueAPI::SMBUS::Disable()
	C#	bool AvalueAPI.SMBUS.Disable()
Unmanaged		bool SMBUS_Disable()

Return Value	Description
FALSE	Class not initialized
TRUE	Success

Remarks:

Call this method if no longer to use SMBUS class methods.

1.6.4 Status

Get status.

Syntax		Description
Managed	C++	AvalueAPI::SMBUS::STATUS AvalueAPI::SMBUS::Status()
	C#	AvalueAPI.SMBUS.STATUS AvalueAPI.SMBUS.Status()
Unmanaged		STATUS SMBUS_Status()

Return Value:

Refer to **1.6.1.1 STATUS**.

1.6.5 Read

Read data.

Syntax		Description
Managed	C++	bool AvalueAPI::SMBUS::Read(AvalueAPI::RWSIZE size, unsigned char addr, unsigned char reg, unsigned long* data)
	C#	bool AvalueAPI.SMBUS.Read(AvalueAPI.RWSIZE size, byte addr, byte reg, uint* data)
Unmanaged		bool SMBUS_Read(RWSIZE size, unsigned char addr, unsigned char reg, unsigned long* data)

Parameters	Description
size	Refer to 1.1.1.2 RWSIZE
addr	Read address
reg	Read register
data	Return data

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.6.6 Write

Write data.

Syntax		Description
Managed	C++	bool AvalueAPI::SMBUS::Write(AvalueAPI::RWSIZE size, unsigned char addr, unsigned char reg, unsigned long data)
	C#	bool AvalueAPI.SMBUS.Write(AvalueAPI.RWSIZE size, byte addr, byte reg, uint data)
Unmanaged		bool SMBUS_Write(RWSIZE size, unsigned char addr, unsigned char reg, unsigned long data)

Parameters	Description
size	Refer to 1.1.1.2 RWSIZE
addr	Write address
reg	Write register
data	Write data

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.6.7 I2C

1.6.7.1 Enable

Change the formatting of some commands to communicate with I2C devices.

Syntax		Description
Managed	C++	bool AvalueAPI::SMBUS::I2C_Enable()
	C#	bool AvalueAPI.SMBUS.I2C_Enable()
Unmanaged		bool SMBUS_I2C_Enable()

Return Value	Description
FALSE	Class not initialized or method not supported
TRUE	Success

1.6.7.2 Disable

Change the formatting of some commands back to SMBUS.

Syntax		Description
Managed	C++	bool AvalueAPI::SMBUS::I2C_Disable()
	C#	bool AvalueAPI.SMBUS.I2C_Disable()
Unmanaged		bool SMBUS_I2C_Disable()

Return Value	Description
FALSE	Class not initialized or method not supported
TRUE	Success

1.7 EC

Embed controller.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::EC
	C#	Avalue.AvalueAPI.EC

1.7.1 Init

1.7.1.1 Default

EC class initialize.

Syntax		Description
Managed	C++	bool AvalueAPI::EC::Init()
	C#	bool AvalueAPI.EC.Init()
Unmanaged		bool EC_Init_Default()

Return Value	Description
FALSE	API not started or class not supported
TRUE	Success

Remarks:

Call this method first before use EC class methods.

1.7.1.2 Select

EC class initialize with ID.

Syntax		Description
Managed	C++	bool AvalueAPI::EC::Init(unsigned short id)
	C#	bool AvalueAPI.EC.Init(ushort id)
Unmanaged		bool EC_Init(unsigned short id)

Parameters	Description
id	Device ID (hex)

Return Value	Description
FALSE	API not started or class not supported
TRUE	Success

Remarks:

Call this method first before use EC class methods.

1.7.2 Disable

EC class disable.

Syntax		Description
Managed	C++	bool AvalueAPI::EC::Disable()
	C#	bool AvalueAPI.EC.Disable()
Unmanaged		bool EC_Disable()

Return Value	Description
FALSE	Class not initialized
TRUE	Success

Remarks:

Call this method if no longer to use EC class methods.

1.7.3 ID

Get the initialized EC device ID.

Syntax		Description
Managed	C++	unsigned short AvalueAPI::EC::ID()
	C#	ushort AvalueAPI.EC.ID()
Unmanaged		unsigned short EC_ID()

Return Value	Description
0xFFFF	Class not initialized or method not supported
Others	Device ID (hex)

1.7.4 BRAM

1.7.4.1 Read

Read EC BRAM data.

Syntax		Description
Managed	C++	unsigned char AvalueAPI::EC::BRAM_Read(unsigned char reg)
	C#	byte AvalueAPI.EC.BRAM_Read(byte reg)
Unmanaged		unsigned char EC_BRAM_Read(unsigned char reg)

Parameters	Description
reg	BRAM register

Return Value	Description
0xFF	Class not initialized or method not supported
0x00 ~ 0xFF	Return data

1.7.4.2 Write

Write EC BRAM data.

Syntax		Description
Managed	C++	bool AvalueAPI::EC::BRAM_Write(unsigned char reg, unsigned char data)
	C#	bool AvalueAPI.EC.BRAM_Write(byte reg, byte data)
Unmanaged		bool EC_BRAM_Write(unsigned char reg, unsigned char data)

Parameters	Description
reg	BRAM register
data	Write data

Return Value	Description
FALSE	Class not initialized or method not supported
TRUE	Success

1.7.5 SRAM

1.7.5.1 Read

Read EC SRAM data.

Syntax		Description
Managed	C++	unsigned char AvalueAPI::EC::SRAM_Read(unsigned short addr)
	C#	byte AvalueAPI.EC.SRAM_Read(ushort addr)
Unmanaged		unsigned char EC_SRAM_Read(unsigned short addr)

Parameters	Description
addr	SRAM address

Return Value	Description
0xFF	Class not initialized or method not supported
0x00 ~ 0xFF	Return data

1.7.5.2 Write

Write EC SRAM data.

Syntax		Description
Managed	C++	bool AvalueAPI::EC::SRAM_Write(unsigned short addr, unsigned char data)
	C#	bool AvalueAPI.EC.SRAM_Write(ushort addr, byte data)
Unmanaged		bool EC_SRAM_Write(unsigned short addr, unsigned char data)

Parameters	Description
addr	SRAM address
data	Write data

Return Value	Description
FALSE	Class not initialized or method not supported
TRUE	Success

1.8 SIO

Super I/O.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::SIO
	C#	Avalue.AvalueAPI.SIO

1.8.1 Init

SIO class initialize.

Syntax		Description
Managed	C++	bool AvalueAPI::SIO::Init()
	C#	bool AvalueAPI.SIO.Init()
Unmanaged		bool SIO_Init()

Return Value	Description
FALSE	API not started or class not supported
TRUE	Success

Remarks:

Call this method first before use SIO class methods.

1.8.2 Disable

SIO class disable.

Syntax		Description
Managed	C++	bool AvalueAPI::SIO::Disable()
	C#	bool AvalueAPI.SIO.Disable()
Unmanaged		bool SIO_Disable()

Return Value	Description
FALSE	Class not initialized
TRUE	Success

Remarks:

Call this method if no longer to use SIO class methods.

1.8.3 GPIO

1.8.3.1 Init

GPIO initialize.

Syntax		Description
Managed	C++	bool AvalueAPI::SIO::GPIO_Init()
	C#	bool AvalueAPI.SIO.GPIO_Init()
Unmanaged		bool SIO_GPIO_Init()

Return Value	Description
FALSE	Class not initialized or method not supported
TRUE	Success

Remarks:

Call this method first before use SIO GPIO methods.

1.8.3.2 Read

1.8.3.2.1 Input

Read GPI value.

Syntax		Description
Managed	C++	unsigned char AvalueAPI::SIO::GPI_Read()
	C#	byte AvalueAPI.SIO.GPI_Read()
Unmanaged		unsigned char SIO_GPI_Read()

Return Value	Description
0xFF	Class not initialized or method not supported
0x00 ~ 0xFF	Return data

1.8.3.2.2 Output

Read GPO value.

Syntax		Description
Managed	C++	unsigned char AvalueAPI::SIO::GPO_Read()
	C#	byte AvalueAPI.SIO.GPO_Read()
Unmanaged		unsigned char SIO_GPO_Read()

Return Value	Description
0xFF	Class not initialized or method not supported
0x00 ~ 0xFF	Return data

1.8.3.3 Write

Write GPO value.

Syntax		Description
Managed	C++	bool AvalueAPI::SIO::GPO_Write(unsigned char data)
	C#	bool AvalueAPI.SIO.GPO_Write(byte data)
Unmanaged		bool SIO_GPO_Write(unsigned char data)

Parameters	Description
data	Write data

Return Value	Description
FALSE	Class not initialized or method not supported
TRUE	Success

1.9 DIO

Digital I/O.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::DIO
	C#	Avalue.AvalueAPI.DIO

1.9.1 Enumerations

1.9.1.1 MAPPING

Mapping type.

Member Name	Value	Description
NA	-1	Class not initialized or method not supported
IO2	2	2 In / 2 Out
IO4	4	4 In / 4 Out
IO6	6	6 In / 6 Out
IO8	8	8 In / 8 Out
IO10	10	10 In / 10 Out (2 I/O extend)
IO12	12	12 In / 12 Out (2 I/O extend)
IO14	14	14 In / 14 Out (2 I/O extend)
IO16	16	16 In / 16 Out (2 I/O extend)

1.9.1.2 VALUE

Value/Direction.

Member Name	Value	Description
NA	-1	Class not initialized or method not supported
LO	0	Low/Output
HI	1	High/Input

1.9.2 Struct

1.9.2.1 DATA

I/O data.

Member Name		Description
C++	C#	
array<AvalueAPI.DIO.VALUE>^ Pin	AvalueAPI.DIO.VALUE[] Pin	Pin data
unsigned char Port	byte Port	Port data

1.9.3 Init

1.9.3.1 No configuration

DIO class initialize.

Syntax		Description
Managed	C++	bool AvalueAPI::DIO::Init()
	C#	bool AvalueAPI.DIO.Init()
Unmanaged		bool DIO_Init()

Return Value	Description
FALSE	API not started or class not supported
TRUE	Success

Remarks:

Call this method first before use DIO class methods.

1.9.3.2 Default configuration

DIO class initialize and set default I/O direction.

Syntax		Description
Managed	C++	bool AvalueAPI::DIO::Init_Default()
	C#	bool AvalueAP.DIO.Init_Default()
Unmanaged		bool DIO_Init_Default()

Return Value	Description
FALSE	API not started or class not supported
TRUE	Success

Remarks:

Call this method first before use DIO class methods.

1.9.4 Disable

DIO class disable.

Syntax		Description
Managed	C++	bool AvalueAPI::DIO::Disable()
	C#	bool AvalueAPI.DIO.Disable()
Unmanaged		bool DIO_Disable()

Return Value	Description
FALSE	Class not initialized
TRUE	Success

Remarks:

Call this method if no longer to use DIO class methods.

1.9.5 Mapping Type

Get mapping type.

Syntax		Description
Managed	C++	AvalueAPI::DIO::MAPPING AvalueAPI::DIO::MappingType()
	C#	AvalueAPI.DIO.MAPPING AvalueAPI.DIO.MappingType()
Unmanaged		MAPPING DIO_MappingType()

Return Value:

Refer to [1.9.1.1 MAPPING](#).

1.9.6 Config

1.9.6.1 Get

Get direction.

Syntax		Description
Managed	C++	AvalueAPI::DIO::DATA AvalueAPI::DIO::Config(int device, int port)
	C#	AvalueAPI.DIO.DATA AvalueAPI.DIO.Config(int device, int port)
Unmanaged		unsigned char DIO_Config_Get(int device, int port)

Parameters	Description
device	0 ~ 1
port	0 ~ 1

Return Value:

Refer to [1.9.2.1 DATA](#).

1.9.6.2 Set

Set direction.

1.9.6.2.1 Port

Syntax		Description
Managed	C++	bool AvalueAPI::DIO::Config(int device, int port, unsigned char data)
	C#	bool AvalueAPI.DIO.Config(int device, int port, byte data)
Unmanaged		bool DIO_Config_Set(int device, int port, unsigned char data)

Parameters	Description
device	0 ~ 1
port	0 ~ 1
data	Port data

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.9.6.2.2 All pin

Syntax		Description
Managed	C++	bool AvalueAPI::DIO::Config(int device, int port, array< AvalueAPI::DIO::VALUE>^ pin)
	C#	bool AvalueAPI.DIO.Config(int device, int port, AvalueAPI.DIO.VALUE[] pin)

Parameters	Description
device	0 ~ 1
port	0 ~ 1
pin	All pin value Refer to 1.9.1.2 VALUE

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.9.6.2.3 Pin

Syntax		Description
Managed	C++	bool AvalueAPI::DIO::Config(int device, int port, int pin, AvalueAPI::DIO::VALUE value)
	C#	bool AvalueAPI.DIO.Config(int device, int port, int pin, AvalueAPI.DIO.VALUE value)

Parameters	Description
device	0 ~ 1
port	0 ~ 1
pin	Pin number
value	Refer to 1.9.1.2 VALUE

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.9.7 Read

Read status.

Syntax		Description
Managed	C++	AvalueAPI::DIO::DATA AvalueAPI::DIO::Read(int device, int port)
	C#	AvalueAPI.DIO.DATA AvalueAPI.DIO.Read(int device, int port)
Unmanaged		unsigned char DIO_Read(int device, int port)

Parameters	Description
device	0 ~ 1
port	0 ~ 1

Return Value:

Refer to [1.9.2.1 DATA](#).

1.9.8 Write

Write status.

1.9.8.1 Port

Syntax		Description
Managed	C++	bool AvalueAPI::DIO::Write(int device, int port, unsigned char data)
	C#	bool AvalueAPI.DIO.Write(int device, int port, byte data)
Unmanaged		bool DIO_Write(int device, int port, unsigned char data)

Parameters	Description
device	0 ~ 1
port	0 ~ 1
data	Port data

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.9.8.2 All pin

Syntax		Description
Managed	C++	bool AvalueAPI::DIO::Write(int device, int port, array< AvalueAPI::DIO::VALUE>^ pin)
	C#	bool AvalueAPI.DIO.Write(int device, int port, AvalueAPI.DIO.VALUE[] pin)

Parameters	Description
device	0 ~ 1
port	0 ~ 1
pin	All pin value Refer to 1.9.1.2 VALUE

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.9.8.3 Pin

Syntax		Description
Managed	C++	bool AvalueAPI::DIO::Write(int device, int port, int pin, AvalueAPI::DIO::VALUE value)
	C#	bool AvalueAPI.DIO.Write(int device, int port, int pin, AvalueAPI.DIO.VALUE value)

Parameters	Description
device	0 ~ 1
port	0 ~ 1
pin	Pin number
value	Refer to 1.9.1.2 VALUE

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.10 WDT

Watchdog timer.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::WDT
	C#	Avalue.AvalueAPI.WDT

1.10.1 Enumerations

1.10.1.1 UNIT

Countdown unit.

Member Name	Value	Description
SECOND	0	Second
MINUTE	1	Minute

1.10.2 Init

Watchdog timer class initialize.

Syntax		Description
Managed	C++	bool AvalueAPI::WDT::Init()
	C#	bool AvalueAPI.WDT.Init()
Unmanaged		bool WDT_Init()

Return Value	Description
FALSE	API not started or class not supported
TRUE	Success

Remarks:

Call this method first before use WDT class methods.

1.10.3 Disable

Watchdog timer class disable.

Syntax		Description
Managed	C++	bool AvalueAPI::WDT::Disable()
	C#	bool AvalueAPI.WDT.Disable()
Unmanaged		bool WDT_Disable()

Return Value	Description
FALSE	Class not initialized
TRUE	Success

Remarks:

Call this method if no longer to use WDT class methods.

1.10.4 Start

Start countdown.

Syntax		Description
Managed	C++	bool AvalueAPI::WDT::Start(AvalueAPI::WDT::UNIT unit, unsigned short time)
	C#	bool AvalueAPI.WDT.Start(AvalueAPI.WDT.UNIT unit, ushort time)
Unmanaged		bool WDT_Start(UNIT unit, unsigned short time)

Parameters	Description
unit	Refer to 1.10.1.1 UNIT
time	SECOND: 0 ~ 65535 MINUTE: 0 ~ 1092

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.10.5 Stop

Stop countdown.

Syntax		Description
Managed	C++	bool AvalueAPI::WDT::Stop()
	C#	bool AvalueAPI.WDT.Stop()
Unmanaged		bool WDT_Stop()

Return Value	Description
FALSE	Class not initialized or method not supported
TRUE	Success

1.10.6 Countdown

Get countdown.

Syntax		Description
Managed	C++	int AvalueAPI::WDT::Countdown()
	C#	int AvalueAPI.WDT.Countdown()
Unmanaged		int WDT_Countdown()

Return Value	Description
-1	Class not initialized or method not supported
Others	Countdown time

1.11 HWM

Hardware monitor.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::HWM
	C#	Avalue.AvalueAPI.HWM

1.11.1 Definition

1.11.1.1 Smart Fan Mode

Name	Value	Description	Description (MX610H / MX610HD)
Mode 0	0	Manual Mode	
Mode 1	1	Mode 1	Target Temperature 40°C. Minimum Fan Speed 0%.
Mode 2	2	Mode 2	Target Temperature 40°C. Minimum Fan Speed 25%.
Mode 3	3	Mode 3	Target Temperature 40°C. Minimum Fan Speed 50%.
Mode 4	4	Mode 4	Target Temperature 40°C. Minimum Fan Speed 75%.
Mode 5	5	Mode 5	Target Temperature 45°C. Minimum Fan Speed 0%.
Mode 6	6	Mode 6	Target Temperature 45°C. Minimum Fan Speed 25%.
Mode 7	7	Mode 7	Target Temperature 45°C. Minimum Fan Speed 50%.
Mode 8	8	Mode 8	Target Temperature 45°C. Minimum Fan Speed 75%.
Mode 9	9	Mode 9	Target Temperature 50°C. Minimum Fan Speed 0%.
Mode 10	10	Mode 10	Target Temperature 50°C. Minimum Fan Speed 25%.
Mode 11	11	Mode 11	Target Temperature 50°C. Minimum Fan Speed 50%.
Mode 12	12	Mode 12	Target Temperature 50°C. Minimum Fan Speed 75%.
Mode 13	13	Mode 13	Target Temperature 55°C. Minimum Fan Speed 0%.
Mode 14	14	Mode 14	Target Temperature 55°C. Minimum Fan Speed 25%.
Mode 15	15	Mode 15	Target Temperature 55°C. Minimum Fan Speed 50%.
Mode 16	16	Mode 16	Target Temperature 55°C. Minimum Fan Speed 75%.
Mode 17	17	Mode 17	Target Temperature 60°C. Minimum Fan Speed 0%.
Mode 18	18	Mode 18	Target Temperature 60°C. Minimum Fan Speed 25%.
Mode 19	19	Mode 19	Target Temperature 60°C. Minimum Fan Speed 50%.
Mode 20	20	Mode 20	Target Temperature 60°C. Minimum Fan Speed 75%.

1.11.2 Enumerations

1.11.2.1 IC

IC controller.

Member Name	Value	Description
EC	0	Embed controller
SIO	1	Super I/O

1.11.3 Init

Hardware monitor class initialize.

Syntax		Description
Managed	C++	bool AvalueAPI::HWM::Init()
	C#	bool AvalueAPI.HWM.Init()
Unmanaged		bool HWM_Init()

Return Value	Description
FALSE	API not started or class not supported
TRUE	Success

Remarks:

Call this method first before use HWM class methods.

1.11.4 Disable

Hardware monitor class disable.

Syntax		Description
Managed	C++	bool AvalueAPI::HWM::Disable()
	C#	bool AvalueAPI.HWM.Disable()
Unmanaged		bool HWM_Disable()

Return Value	Description
FALSE	Class not initialized
TRUE	Success

Remarks:

Call this method if no longer to use HWM class methods.

1.11.5 Temperature

1.11.5.1 CPU

Get CPU temperature.

Syntax		Description
Managed	C++	float AvalueAPI::HWM::Temperature_CPU()
	C#	float AvalueAPI.HWM.Temperature_CPU()
Unmanaged		float HWM_Temperature_CPU()

Return Value	Description
-999	Class not initialized or method not supported
Others	Celsius

1.11.5.2 CPU PECI

Get CPU PECI temperature.

Syntax		Description
Managed	C++	float AvalueAPI::HWM::Temperature_CPU_PECI()
	C#	float HWM_Temperature_CPU_PECI()
Unmanaged		float AvalueAPI.HWM.Temperature_CPU_PECI()

Return Value	Description
-999	Class not initialized or method not supported
Others	Celsius

1.11.5.3 System

Get system temperature.

Syntax		Description
Managed	C++	float AvalueAPI::HWM::Temperature_System()
	C#	float AvalueAPI.HWM.Temperature_System()
Unmanaged		float HWM_Temperature_System()

Return Value	Description
-999	Class not initialized or method not supported
Others	Celsius

1.11.6 Voltage

1.11.6.1 Value

Get voltage.

Syntax		Description
Managed	C++	float AvalueAPI::HWM::Voltage(AvalueAPI::HWM::IC ic, int number)
	C#	float AvalueAPI.HWM.Voltage(AvalueAPI.HWM.IC ic, int number)
Unmanaged		float HWM_Voltage(IC ic, int number)

Parameters	Description
ic	Refer to 1.11.2.1 IC
number	EC: 0 ~ 4 SIO: 0 ~ 7

Return Value	Description
-1	Class not initialized, method not supported or parameters incorrect
Others	Voltage

1.11.6.2 Text

Get voltage text.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::HWM::Voltage_Text(AvalueAPI::HWM::IC ic, int number)
	C#	string AvalueAPI.HWM.Voltage_Text(AvalueAPI.HWM.IC ic, int number)
Unmanaged		wchar_t* HWM_Voltage_Text(IC ic, int number)

Parameters	Description
ic	Refer to 1.11.2.1 IC
number	EC: 0 ~ 4 SIO: 0 ~ 7

Return Value	Description
N/A	Class not initialized, method not supported or parameters incorrect
Others	Voltage text

1.11.7 PWM

1.11.7.1 Get

Get PWM.

Syntax		Description
Managed	C++	int AvalueAPI::HWM::PWM(int number)
	C#	int AvalueAPI.HWM.PWM(int number)
Unmanaged		int HWM_PWM_Get(int number)

Parameters	Description
number	0 ~ 5

Return Value	Description
-1	Class not initialized, method not supported or parameters incorrect
0 ~ 255	PWM value

1.11.7.2 Set

Set PWM.

Syntax		Description
Managed	C++	bool AvalueAPI::HWM::PWM(int number, unsigned char value)
	C#	bool AvalueAPI.HWM.PWM(int number, byte value)
Unmanaged		bool HWM_PWM_Set(int number, unsigned char value)

Parameters	Description
number	0 ~ 5
value	0x00 ~ 0xFF

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.11.7.3 Text

Get PWM text.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::HWM::PWM_Text(int number)
	C#	string AvalueAPI.HWM.PWM_Text(int number)
Unmanaged		wchar_t* HWM_PWM_Text(int number)

Parameters	Description
number	0 ~ 5

Return Value	Description
N/A	Class not initialized, method not supported or parameters incorrect
Others	PWM text

1.11.8 Fan

1.11.8.1 Get Mode

Get smart fan mode.

Syntax		Description
Managed	C++	int AvalueAPI::HWM::Fan_Mode(int number)
	C#	int AvalueAPI.HWM.Fan_Mode(int number)
Unmanaged		int HWM_Fan_Mode_Get(int number)

Parameters	Description
number	0 ~ 1

Return Value	Description
-1	Class not initialized, method not supported or parameters incorrect
Others	See 1.11.1.1 Smart Fan Mode

1.11.8.2 Set Mode

Set smart fan mode.

Syntax		Description
Managed	C++	bool AvalueAPI::HWM::Fan_Mode(int number, int mode)
	C#	bool AvalueAPI.HWM.Fan_Mode(int number, int mode)
Unmanaged		bool HWM_Fan_Mode_Set(int number, int mode)

Parameters	Description
number	0 ~ 1
mode	See 1.11.1.1 Smart Fan Mode

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.11.8.3 Mode Text

Get smart fan mode text.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::HWM::Fan_Mode_Text(int number)
	C#	string AvalueAPI.HWM.Fan_Mode_Text(int number)
Unmanaged		wchar_t* HWM_Fan_Mode_Text(int number)

Parameters	Description
number	0 ~ 1

Return Value	Description
N/A	Class not initialized, method not supported or parameters incorrect
Others	Smart fan mode text

1.11.8.4 Get Speed

Get fan speed.

Syntax		Description
Managed	C++	int AvalueAPI::HWM::Fan_Speed(int number)
	C#	int AvalueAPI.HWM.Fan_Speed(int number)
Unmanaged		int HWM_Fan_Speed_Get(int number)

Parameters	Description
number	0 ~ 2

Return Value	Description
-1	Class not initialized, method not supported or parameters incorrect
Others	RPM

1.11.8.5 Get PWM

Get fan PWM.

Syntax		Description
Managed	C++	int AvalueAPI::HWM::Fan_PWM(int number)
	C#	int AvalueAPI.HWM.Fan_PWM(int number)
Unmanaged		int HWM_Fan_PWM_Get(int number)

Parameters	Description
number	0 ~ 2

Return Value	Description
-1	Class not initialized, method not supported or parameters incorrect
0 ~ 255	PWM value

1.11.8.6 Set PWM

Set fan PWM.

Syntax		Description
Managed	C++	bool AvalueAPI::HWM::Fan_PWM(int number, unsigned char value)
	C#	bool AvalueAPI.HWM.Fan_PWM(int number, byte value)
Unmanaged		bool HWM_Fan_PWM_Set(int number, unsigned char value)

Parameters	Description
number	0 ~ 2
value	0x00 ~ 0xFF PWM value

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.11.8.7 Fan Text

Get fan text.

Syntax		Description
Managed	C++	System::String^ AvalueAPI::HWM::Fan_Text(int number)
	C#	string AvalueAPI.HWM.Fan_Text(int number)
Unmanaged		wchar_t* HWM_Fan_Text(int number)

Parameters	Description
number	0 ~ 2

Return Value	Description
N/A	Class not initialized, method not supported or parameters incorrect
Others	Fan text

1.12 LCD

LVDS LCD panel.

Syntax		Description
Managed	C++	Avalue::AvalueAPI::LCD
	C#	Avalue.AvalueAPI.LCD

1.12.1 Enumerations

1.12.1.1 SPEC

Specifications.

Member Name	Value	Description	Description (MX610H)	Description (MX610HD)
NA	-1	Class not initialized or method not supported		
R1024X768_B24_C1	0	1024x768, 24/1	800x600, 18	800x600 18/1
R800X600_B18_C1	1	800x600, 18/1	1024x768, 18	1024x768 18/1
R1024X768_B18_C1	2	1024x768, 18/1	1024x768, 24	1024x768 24/1
R1366X768_B18_C1	3	1366x768, 18/1	1280x768, 18	1280x768 18/1
R1024X600_B18_C1	4	1024x600, 18/1	1280x800, 18	1280x800 24/1
R1280X800_B18_C1	5	1280x800, 18/1	1280x960, 18	1280x960 18/1
R1920X1200_B24_C2	6	1920x1200, 24/2	1280x1024, 24	1280x1024 24/2
R640X480_B18_C1	7	640x480, 18/1	1366x768, 18	1366x768 18/1
R800X480_B18_C1	8	800x480, 18/1	1366x768, 24	1366x768 24/1
R1920X1080_B18_C2	9	1920x1080, 18/2	1440x900, 24	1440x900 24/2
R1280X1024_B24_C2	10	1280x1024, 24/2	1400x1050, 24	1400x1050 24/2
R1440X900_B18_C2	11	1440x900, 18/2	1600x900, 24	1600x900 24/2
R1600X1200_B24_C2	12	1600x1200, 24/2	1680x1050, 24	1680x1050 24/2
R1366X768_B24_C1	13	1366x768, 24/1	1600x1200, 24	800x480 24/1
R1920X1080_B24_C2	14	1920x1080, 24/2	1920x1080, 24	1920x1080 24/2
R1680X1050_B24_C2	15	1680x1050, 24/2	1920x1200, 24	1920x1200 24/2

1.12.1.2 CLOCK

Clock frequency.

Member Name	Value	Description	Description (MX610HD)
NA	-1	Class not initialized or method not supported	
F200	0	200 Hz	200 Hz
F300	1	300 Hz	250 Hz
F400	2	400 Hz	333 Hz
F500	3	500 Hz	500 Hz
F700	4	700 Hz	1 kHz
F1K	5	1 KHz	11.6 kHz
F2K	6	2 KHz	15.6 kHz
F3K	7	3 KHz	23.3 kHz
F5K	8	5 KHz	Not supported
F10K	9	10 KHz	Not supported
F20K	10	20 KHz	Not supported

1.12.1.3 CTRL

Control mode.

Member Name	Value	Description
NA	-1	Class not initialized or method not supported
BIOS	0	BIOS control
BUTTON	1	Button control
VR	2	Variable resistance control
OS	3	OS brightness bar control

1.12.1.4 LIGHT

Backlight status.

Member Name	Value	Description
NA	-1	Class not initialized or method not supported
OFF	0	Off
ON	1	On

1.12.2 Init

LCD class initialize.

Syntax		Description
Managed	C++	bool AvalueAPI::LCD::Init()
	C#	bool AvalueAPI.LCD.Init()
Unmanaged		bool LCD_Init()

Return Value	Description
FALSE	API not started or class not supported
TRUE	Success

Remarks:

Call this method first before use LCD class methods.

1.12.3 Disable

LCD class disable.

Syntax		Description
Managed	C++	bool AvalueAPI::LCD::Disable()
	C#	bool AvalueAPI.LCD.Disable()
Unmanaged		bool LCD_Disable()

Return Value	Description
FALSE	Class not initialized
TRUE	Success

Remarks:

Call this method if no longer to use LCD class methods.

1.12.4 Type

Get type specifications.

Syntax		Description
Managed	C++	AvalueAPI::LCD::SPEC AvalueAPI::LCD::Type()
	C#	AvalueAPI.LCD.SPEC AvalueAPI.LCD.Type()
Unmanaged		SPEC LCD_Type()

Return Value:

Refer to [1.12.1.1 SPEC](#).

1.12.5 Clock

1.12.5.1 Get

Get clock frequency.

Syntax		Description
Managed	C++	AvalueAPI::LCD::CLOCK AvalueAPI::LCD::Clock()
	C#	AvalueAPI.LCD.CLOCK AvalueAPI.LCD.Clock()
Unmanaged		CLOCK LCD_Clock_Get()

Return Value:

Refer to [1.12.1.2 CLOCK](#).

1.12.5.2 Set

Set clock frequency.

Syntax		Description
Managed	C++	bool AvalueAPI::LCD::Clock(AvalueAPI::LCD::CLOCK clock)
	C#	bool AvalueAPI.LCD.Clock(AvalueAPI.LCD.CLOCK clock)
Unmanaged		bool LCD_Clock_Set(CLOCK clock)

Parameters	Description
clock	Refer to 1.12.1.2 CLOCK

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.12.6 Control

1.12.6.1 Get

Get control mode.

Syntax		Description
Managed	C++	AvalueAPI::LCD::CTRL AvalueAPI::LCD::Control()
	C#	AvalueAPI.LCD.CTRL AvalueAPI.LCD.Control()
Unmanaged		CTRL LCD_Control_Get()

Return Value:

Refer to [1.12.1.3 CTRL](#).

1.12.6.2 Set

Set control mode.

Syntax		Description
Managed	C++	bool AvalueAPI::LCD::Control(AvalueAPI::LCD::CTRL ctrl)
	C#	bool AvalueAPI.LCD.Control(AvalueAPI.LCD.CTRL ctrl)
Unmanaged		bool LCD_Control_Set(CTRL ctrl)

Parameters	Description
ctrl	Refer to 1.12.1.3 CTRL

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.12.7 Backlight

1.12.7.1 Get

Get backlight status.

Syntax		Description
Managed	C++	AvalueAPI::LCD::LIGHT AvalueAPI::LCD::Backlight()
	C#	AvalueAPI.LCD.LIGHT AvalueAPI.LCD.Backlight()
Unmanaged		LIGHT LCD_Backlight_Get()

Return Value:

Refer to [1.12.1.4 LIGHT](#).

1.12.7.2 Set

Set backlight status.

Syntax		Description
Managed	C++	bool AvalueAPI::LCD::Backlight(AvalueAPI::LCD::LIGHT light)
	C#	bool AvalueAPI.LCD.Backlight(AvalueAPI.LCD.LIGHT light)
Unmanaged		bool LCD_Backlight_Set(LIGHT light)

Parameters	Description
light	Refer to 1.12.1.4 LIGHT .

Return Value	Description
FALSE	Class not initialized, method not supported or parameters incorrect
TRUE	Success

1.12.8 Brightness

1.12.8.1 Get

Get brightness.

Syntax		Description
Managed	C++	int AvalueAPI::LCD::Brightness()
	C#	int AvalueAPI.LCD.Brightness()
Unmanaged		int LCD_Brightness_Get()

Return Value	Description
-1	Class not initialized or method not supported
0 ~ 255	BIOS control: 0 ~ 255 PWM value OS control: 0 ~ 100 percentage

1.12.8.2 Set

Set brightness.

Syntax		Description
Managed	C++	bool AvalueAPI::LCD::Brightness(unsigned char value)
	C#	bool AvalueAPI.LCD.Brightness(byte value)
Unmanaged		bool LCD_Brightness_Set(unsigned char value)

Parameters	Description
value	BIOS control: 0x00 ~ 0xFF PWM value OS control: 0 ~ 100 percentage

Return Value	Description
FALSE	Class not initialized, method not supported or control mode is BUTTON or VR
TRUE	Success

2. Support List

2.1 Bay Trail

1. ACP-BYTPOS
2. ARC-BYT
3. EBM-BYT
4. EBM-BYTS
5. EBM-BYTV(VMS-BYT)
6. ECM-BYT
7. ECM-BYT2
8. EMX-BYT2
9. EMX-BYT3
10. EQM-BYT
11. EQM-BYT2
12. ESM-BYT

2.2 Braswell

1. ECM-BSWA

2.3 Apollo Lake

1. ARC-APL
2. EBM-APL
3. EBM-APLV(VMS-APL)
4. ECM-APL
5. ECM-APL2
6. ECM-APL2-B1
7. EMX-APL
8. EMX-APLP
9. EPX-APLP
10. EQM-APL
11. ESM-APLC
12. ESM-APLM
13. NCM-APL(NUC-APL)

14. RITY153

15. RiVar-1501

2.4 Haswell

1. EBM-QM87U
2. ECM-QM87
3. ECM-QM87R
4. EPI-QM87
5. EPI-QM87R

2.5 Sky Lake

2.5.1 SKL-H

1. ECM-SKLH
2. ESM-SKLH

2.5.2 SKL-S

1. EAX-C236KP
2. EAX-Q170KP
3. EAX-Q170P
4. EMX-H110KP
5. EMX-H110P
6. EMX-Q170B
7. EMX-Q170KP
8. EMX-Q170P
9. ERX-C236KP
10. ERX-H110KP
11. ERX-H110P
12. SLP-SKL

2.5.3 SKL-U

1. ARC-SKLU
2. EBM-SKLU
3. EBM-SKLUS(EMS-SKLU)
4. ECM-SKLU

5. EMX-SKLGP
6. EMX-SKLU
7. ESM-SKLU
8. HID-2432

2.6 Kaby Lake

1. ECM-KBLH
2. EMX-KBLU2P
3. ESM-KBLA
4. ESM-KBLH
5. ESM-KBLU
6. HID-2334

2.7 Coffee Lake

1. EAX-C246P
2. EAX-C246P-B1
3. EBM-CFSV(VMS-CFS)
4. ECM-CFS
5. EMX-C246DP
6. EMX-C246P
7. EMX-H310DP
8. EMX-H310P

2.8 Whiskey Lake

1. ECM-WHL
2. EMX-WHLGP

2.9 Comet Lake

1. EAX-W480P
2. ERX-W480P

2.10 Tiger Lake

1. ARC-TGL
2. EBM-TGLS(EMS-TGL)
3. ECM-TGU
4. ECM-TGU-B1(EPC-TGU-B1)
5. ECM-TGUC(ACS10-TGU)
6. EMX-TGLC
7. EMX-TGLC-B1
8. EMX-TGLP
9. ESM-TGH
10. NCM-TGU

2.11 Elkhart Lake

1. ARC-EHL
2. EBM-EHLR(VMS-EHLR)
3. EBM-EHLS(EMS-EHL)
4. ECM-EHL
5. ECM-EHL3
6. EMX-EHLP
7. EQM-EHL
8. ESM-EHLC
9. EZX-EHLP
10. NCM-EHL

2.12 Alder Lake

2.12.1 ADL-N

1. ARC-ADLN
2. NUC-ADN

2.12.2 ADL-P

1. EMX-ADLP

2.12.3 ADL-PS

1. ARC-1242
2. EBM-ADPS(VMS-ADPS)

2.12.4 ADL-S

1. MX610H
2. MX610HD

2.13 Amston Lake

1. ECM-ASL3
2. ESM-ASLC

2.13.1 ASL-N

1. ECM-ASL
2. EMX-ASLP
3. EPX-ASLP

2.14 Raptor Lake

1. ECM-RPL
2. EMX-R680P
3. EMX-RPLP
4. ESM-RPL
5. ESM-RPLC

2.14.1 RPL-S

1. ESM-HRPL

2.14.2 RPL-U

1. NUC-RPU

2.15 Meteor Lake

1. ECM-MTL
2. EMX-MTLP

2.15.1 MTL-H

1. EMS-MTH

2.15.2 MTL-U

1. EMS-MTU

2.16 Arrow Lake

1. ECM-ARL
2. ESM-ARL

2.16.1 ARL-H

1. EMS-ARH

2.16.2 ARL-S

1. EMX-W880P

2.17 Twin Lake

1. ECM-TWL
2. ECM-TWL3
3. EMX-TWLP
4. ESM-TWLC
5. RIVAR-TWL(RiVar-1539-B1)

2.18 Bartlett Lake

2.18.1 BTL-S

1. EAX-R680BP

2.19 Panther Lake

1. ECM-PTL
2. EMX-PTLP
3. ESM-PTL

2.20 Wildcat Lake

1. ECM-WCL(EPC-WCL)

2.21 Zhaoxin KX-6000

1. EMX-ZXEDP

2.22 Zhaoxin KX-6000G

1. EMX-KX60G
2. ESM-KX60G
3. ESM-KX60GC

3. Revision History

Revision	Date	Description
3.00.00.00	2018/01/11	Initial version. Support APL.
3.00.00.01	2018/01/18	Modify function.
3.00.00.02	2018/01/26	Modify function.
3.00.00.03	2018/03/21	Add Information, Setting, LVDS function.
3.00.00.04	2018/03/31	Add Information, HWM function. Modify Configuration file. Modify Setting, SMBUS function.
3.00.01.00	2018/06/06	Support BYT. Add Information, Setting, SMBUS, DIO function. Modify SMBUS function.
3.00.02.00	2018/07/31	Class / Function / Parameter rename. Function / Parameter redefine. Support SKL. Add Super IO function. Modify Configuration file. Modify Information, Setting, SMBUS, LVDS function.
3.00.02.01	2018/12/22	Modify Configuration file. Modify Setting, EC, DIO, HWM, Super IO function.
3.00.02.02	2019/04/10	Modify Configuration file. Modify LVDS function.
3.00.03.00	2019/05/20	Support KBL. Add VMS, HWM function. Modify Configuration file. Modify Setting, WDT, HWM function.
3.00.04.00	2019/07/19	Support CFS. Modify Configuration file. Modify Setting, WDT function.
3.00.04.01	2019/11/15	Modify Configuration file. Modify AvalueAPI, Super IO function.
3.00.04.02	2019/12/31	Modify Configuration file.
3.00.04.03	2020/05/28	Add AvalueAPI, Info, Setting, SMBUS, EC unmanaged function.
3.00.04.04	2020/07/17	Modify Configuration file.

		Modify SMBUS, DIO, WDT, HWM function.
3.00.05.00	2020/12/30	Class / Function / Parameter rename or redefine. Support TGL. Add SIO, DIO, WDT, HWM, LCD unmanaged function. Modify Configuration file. Modify AvalueAPI, Info, Setting, SMBUS, EC unmanaged function.
3.00.05.01	2021/07/20	Modify Configuration file.
3.00.05.02	2022/01/13	Modify Configuration file. Modify EC, DIO function.
3.00.05.03	2022/03/23	Modify Configuration file. Add Info function. Modify Info, Setting, HWM, LCD function.
3.00.05.04	2022/04/08	Modify Configuration file. Modify Info function.
3.00.05.05	2022/06/10	Modify Configuration file. AvalueAPI unmanaged function rename. Modify Setting, EC, SIO, HWM, WDT function.
3.00.05.06	2022/07/22	Modify Configuration file. Add Setting function. Modify EC, HWM function.
3.00.05.07	2022/12/30	Modify Configuration file. Modify HWM function.
3.00.05.08	2023/02/03	Modify Configuration file. Modify Setting, CashDrawer function.
3.00.05.09	2023/04/13	Modify Configuration file. Modify Setting, SMBUS, SIO function.
3.00.05.10	2023/07/20	Modify Configuration file. Add LCD function. Modify EC, HWM function.
3.00.05.11	2023/08/18	Modify Configuration file. Modify HWM function.
3.00.05.12	2023/08/25	Modify Configuration file. Modify WDT function.
3.00.05.13	2023/11/30	Modify Configuration file. Modify Info, SMBUS, DIO function.
3.00.05.14	2023/12/20	Modify Configuration file.

		Modify LCD function.
3.00.05.15	2024/01/15	Modify Configuration file. Modify AvalueAPI, Setting, SIO, DIO, WDT, HWM, LCD function.
3.00.05.16	2024/04/10	Modify Configuration file. Modify DIO, HWM function.
3.00.05.17	2024/05/07	Modify Configuration file. Modify Setting function.
3.00.06.00	2024/05/21	AvalueDriver v3.0.0.0 Modify Configuration file.
3.00.06.01	2024/05/24	Modify Configuration file. Modify DIO function.
3.00.06.02	2024/10/15	Modify Configuration file. Modify Info, LCD function.
3.00.06.03	2024/12/25	Modify Configuration file. Modify Setting, DIO, HWM, function.
3.00.06.04	2025/02/21	Modify Configuration file. Modify DIO function.
3.00.06.05	2025/05/02	Modify Configuration file. Modify Info, Setting, SIO, DIO, WDT, HWM, LCD function.
3.00.06.06	2025/06/03	Modify Configuration file. Modify Setting, EC function.
3.00.06.07	2025/06/06	Modify Info function.
3.00.06.08	2025/06/12	Modify Configuration file. Add Info function. Modify AvalueAPI, Setting function.
3.00.06.09	2025/09/08	Add Info function. Modify DIO function.
3.00.06.10	2025/11/25	Modify Configuration file. Add Info function. Modify Info function.
3.00.06.11	2025/12/08	Modify Configuration file. Modify Info function.
3.00.06.12	2026/03/24	Modify Configuration file. Modify AvalueAPI, Info, EC, DIO, WDT, HWM, LCD function.
3.00.06.13	2026/04/07	Modify AvalueAPI, Info, EC, DIO, SMBUS function.
3.00.06.14	2026/06/26	Modify Configuration file.



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		Modify AvalueAPI, SMBUS function.
3.00.06.15	2026/08/24	Modify Configuration file. Modify Setting, SIO, WDT, HWM, LCD function.
3.00.06.16	2026/09/18	Modify AvalueAPI function. Add Info function.